

# Natural Language Processing

## Lab 3

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# Today's Agenda

1. Terminology
2. Vector Representations
3. TF-IDF Weights
4. Make and Test a Search Engine! (Jupyter Notebook)

# Terminology

'Algorithmic bias describes systematic and repeatable errors in a ...'

'Bias can emerge due to many factors, including but not limited ...'

'Algorithmic bias is found across platforms, including but not ...'

'The study of algorithmic bias is most concerned with algorithms ...'

# Terminology

corpus

'Algorithmic bias describes systematic and repeatable errors in a ...'

'Bias can emerge due to many factors, including but not limited ...'

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# Terminology

document

'Algorithmic bias describes systematic and repeatable errors in a ...'

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# Terminology

'Algorithmic **bias** describes systematic and repeatable errors in a ...'

'Bias can emerge due to many factors, including but not limited ...'

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# Terminology

token, term

'Algorithmic **bias** describes systematic and repeatable errors in a ...'

'Bias can emerge due to many factors, including but not limited ...'

'Algorithmic bias is found across platforms, including but not ...'

'The study of algorithmic bias is most concerned with algorithms ...'

# Terminology

token, term → word, N-gram, subword, ...

'Algorithmic **bias** describes systematic and repeatable errors in a ...'

'Bias can emerge due to many factors, including but not limited ...'

'Algorithmic bias is found across platforms, including but not ...'

'The study of algorithmic bias is most concerned with algorithms ...'

# Terminology

doc1

doc2

doc3

doc4

| across | algorithm | algorithmic | and | arbitrary | bias | ... |
|--------|-----------|-------------|-----|-----------|------|-----|
|        |           |             |     |           |      |     |

# Terminology

doc1

doc2

doc3

doc4

vocabulary, lexicon (ordered)

| across | algorithm | algorithmic | and | arbitrary | bias | ... |
|--------|-----------|-------------|-----|-----------|------|-----|
|        |           |             |     |           |      |     |

# Terminology

doc1

doc2

doc3

doc4

vocabulary, lexicon (ordered) → feature, dimension

| across | algorithm | algorithmic | and | arbitrary | bias | ... |
|--------|-----------|-------------|-----|-----------|------|-----|
|        |           |             |     |           |      |     |

# Terminology

doc1

doc2

doc3

doc4

| across | algorithm | algorithmic | and | arbitrary | bias | ... |
|--------|-----------|-------------|-----|-----------|------|-----|
|        |           |             |     |           |      |     |

# Terminology

doc1

doc2

doc3

doc4

| across | algorithm | algorithmic | and | arbitrary | bias | ... |
|--------|-----------|-------------|-----|-----------|------|-----|
|        |           |             |     |           |      |     |

vector (numerical)

# Terminology

doc1

doc2

doc3

doc4

| across | algorithm | algorithmic | and | arbitrary | bias | ... |
|--------|-----------|-------------|-----|-----------|------|-----|
|        |           |             |     |           |      |     |

Vector Space Model (VSM)

# Represent. - One-Hot Encoding

'Algorithmic bias describes systematic and repeatable errors in a ...'

|             | across | algorithm | algorithmic | and | arbitrary | bias | ... |
|-------------|--------|-----------|-------------|-----|-----------|------|-----|
| algorithmic | 0      | 0         | 1           | 0   | 0         | 0    |     |
| bias        | 0      | 0         | 0           | 0   | 0         | 1    |     |
| describes   | 0      | 0         | 0           | 0   | 0         | 0    |     |
| systematic  | 0      | 0         | 0           | 0   | 0         | 0    |     |
| and         | 0      | 0         | 0           | 1   | 0         | 0    |     |

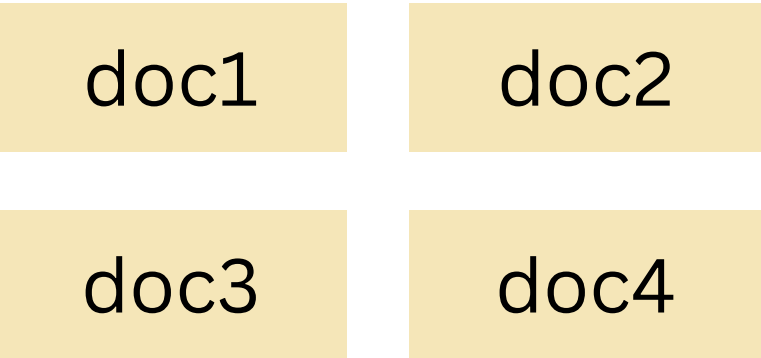
# Represent. - One-Hot Encoding

✓ retain sequence  
✗ super sparse

'Algorithmic bias describes systematic and repeatable errors in a ...'

|             | across | algorithm | algorithmic | and | arbitrary | bias | ... |
|-------------|--------|-----------|-------------|-----|-----------|------|-----|
| algorithmic | 0      | 0         | 1           | 0   | 0         | 0    |     |
| bias        | 0      | 0         | 0           | 0   | 0         | 1    |     |
| describes   | 0      | 0         | 0           | 0   | 0         | 0    |     |
| systematic  | 0      | 0         | 0           | 0   | 0         | 0    |     |
| and         | 0      | 0         | 0           | 1   | 0         | 0    |     |

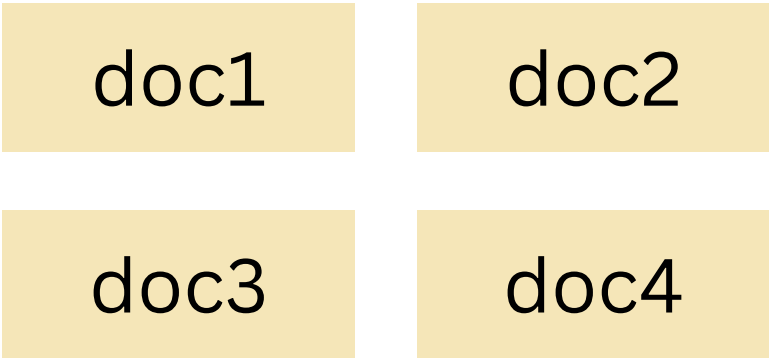
# Represent. - Bag of Words



|       | across | algorithm | algorithmic | and | arbitrary | bias | ... |
|-------|--------|-----------|-------------|-----|-----------|------|-----|
| doc 1 | 0      | 0         | 1           | 1   | 1         | 1    |     |
| doc 2 | 0      | 2         | 0           | 0   | 0         | 1    |     |
| doc 3 | 0      | 0         | 1           | 3   | 0         | 1    |     |
| doc 4 | 0      | 0         | 1           | 1   | 0         | 1    |     |

# Represent. - Bag of Words

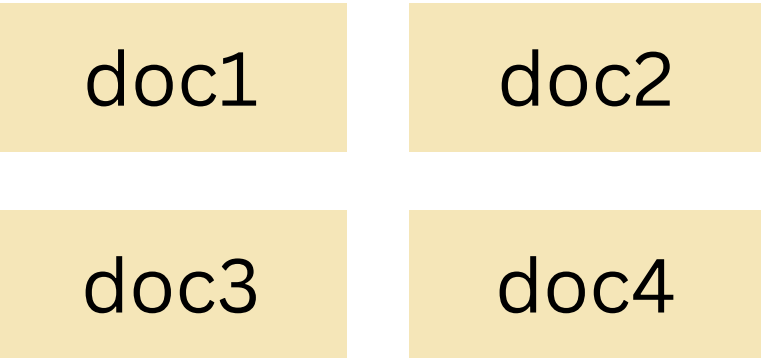
✓ squeeze all docs  
✗ lose sequence



|       | across | algorithm | algorithmic | and | arbitrary | bias | ... |
|-------|--------|-----------|-------------|-----|-----------|------|-----|
| doc 1 | 0      | 0         | 1           | 1   | 1         | 1    |     |
| doc 2 | 0      | 2         | 0           | 0   | 0         | 1    |     |
| doc 3 | 0      | 0         | 1           | 3   | 0         | 1    |     |
| doc 4 | 0      | 0         | 1           | 1   | 0         | 1    |     |

# Represent. - Bag of Words

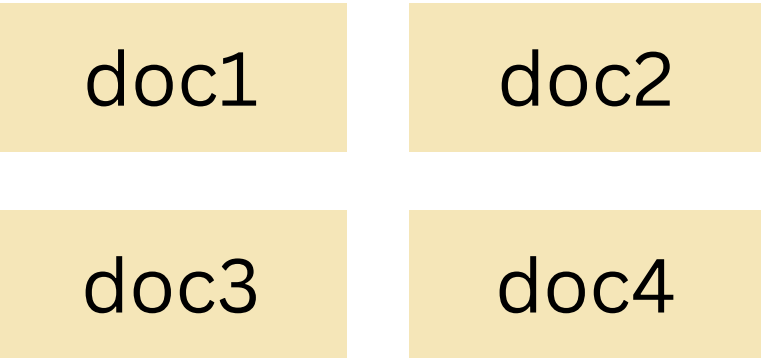
✓ squeeze all docs  
✗ lose sequence



document-term matrix

|       | across | algorithm | algorithmic | and | arbitrary | bias | ... |
|-------|--------|-----------|-------------|-----|-----------|------|-----|
| doc 1 | 0      | 0         | 1           | 1   | 1         | 1    |     |
| doc 2 | 0      | 2         | 0           | 0   | 0         | 1    |     |
| doc 3 | 0      | 0         | 1           | 3   | 0         | 1    |     |
| doc 4 | 0      | 0         | 1           | 1   | 0         | 1    |     |

# Represent. - Bag of N-Grams



|       | across | across<br>platforms | algorithm | algorithm<br>or | algorithmic | algorithmic<br>bias | ... |
|-------|--------|---------------------|-----------|-----------------|-------------|---------------------|-----|
| doc 1 | 0      | 0                   | 0         | 0               | 1           | 1                   |     |
| doc 2 | 0      | 0                   | 2         | 1               | 0           | 0                   |     |
| doc 3 | 1      | 1                   | 0         | 0               | 1           | 1                   |     |
| doc 4 | 0      | 0                   | 0         | 0               | 1           | 1                   |     |

# Represent. - Bag of N-Grams

✓ neighboring sequence  
✗ vocabulary size increase

doc1    doc2

doc3    doc4

|       | across | across<br>platforms | algorithm | algorithm<br>or | algorithmic | algorithmic<br>bias | ... |
|-------|--------|---------------------|-----------|-----------------|-------------|---------------------|-----|
| doc 1 | 0      | 0                   | 0         | 0               | 1           | 1                   |     |
| doc 2 | 0      | 0                   | 2         | 1               | 0           | 0                   |     |
| doc 3 | 1      | 1                   | 0         | 0               | 1           | 1                   |     |
| doc 4 | 0      | 0                   | 0         | 0               | 1           | 1                   |     |

# TF-IDF: Intuition

Course Directory

Close filter ^

Year

2026

Semester

Fall

Study program

MDS - Master of Data Science

Lecturers

Lecturers

Q Search

Q My courses

course with assignments

# TF-IDF: Intuition

Course Directory

Close filter ^

Year

2026

Semester

Fall

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course with assignments

Course Directory

Close filter ^

Year

2026

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Lecturers

Lecturers

Q Search

Q My courses

natural language processing

# TF-IDF: Math

1. **Term Frequency**: how often does the term appear in THIS document?
2. **Inverse Document Frequency**: how rare is the term relative to the entire corpus?
  - a.  $N_d$ : total number of documents
  - b.  $DF(t)$ : number of documents the term appears in

| word      | TF(t, d) | $N_d$ | $DF(t)$ | IDF | TF*IDF |
|-----------|----------|-------|---------|-----|--------|
| or        | 1        | 4     | 4       |     |        |
| biases    | 1        | 4     | 1       |     |        |
| platforms | 2        | 4     | 1       |     |        |

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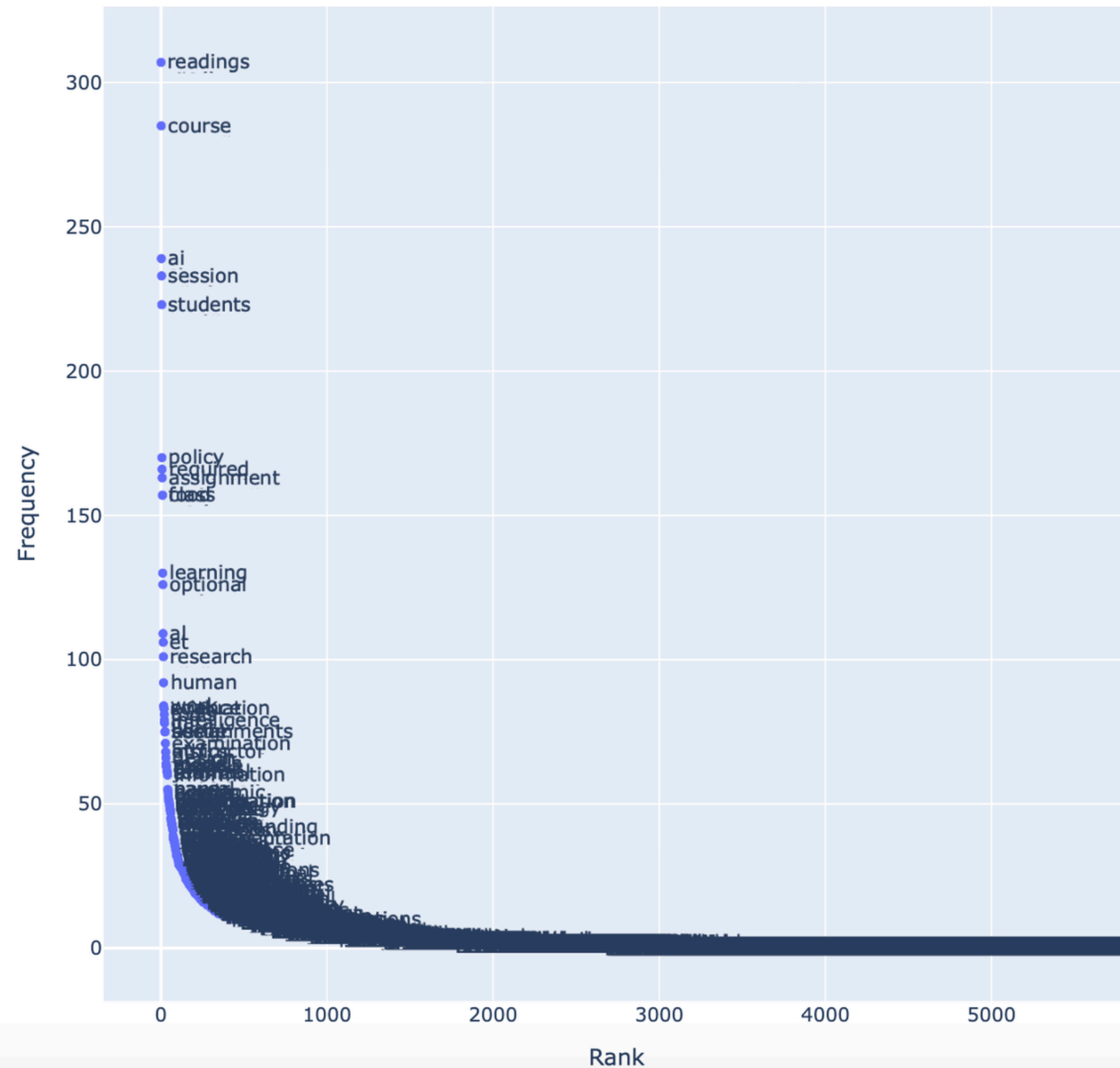
| word      | TF(t, d) | $N_d$ | DF(t) | IDF   | TF*IDF |
|-----------|----------|-------|-------|-------|--------|
| or        | 1        | 4     | 4     | $4/4$ |        |
| biases    | 1        | 4     | 1     | $4/1$ |        |
| platforms | 2        | 4     | 1     | $4/1$ |        |

# TF-IDF: Math

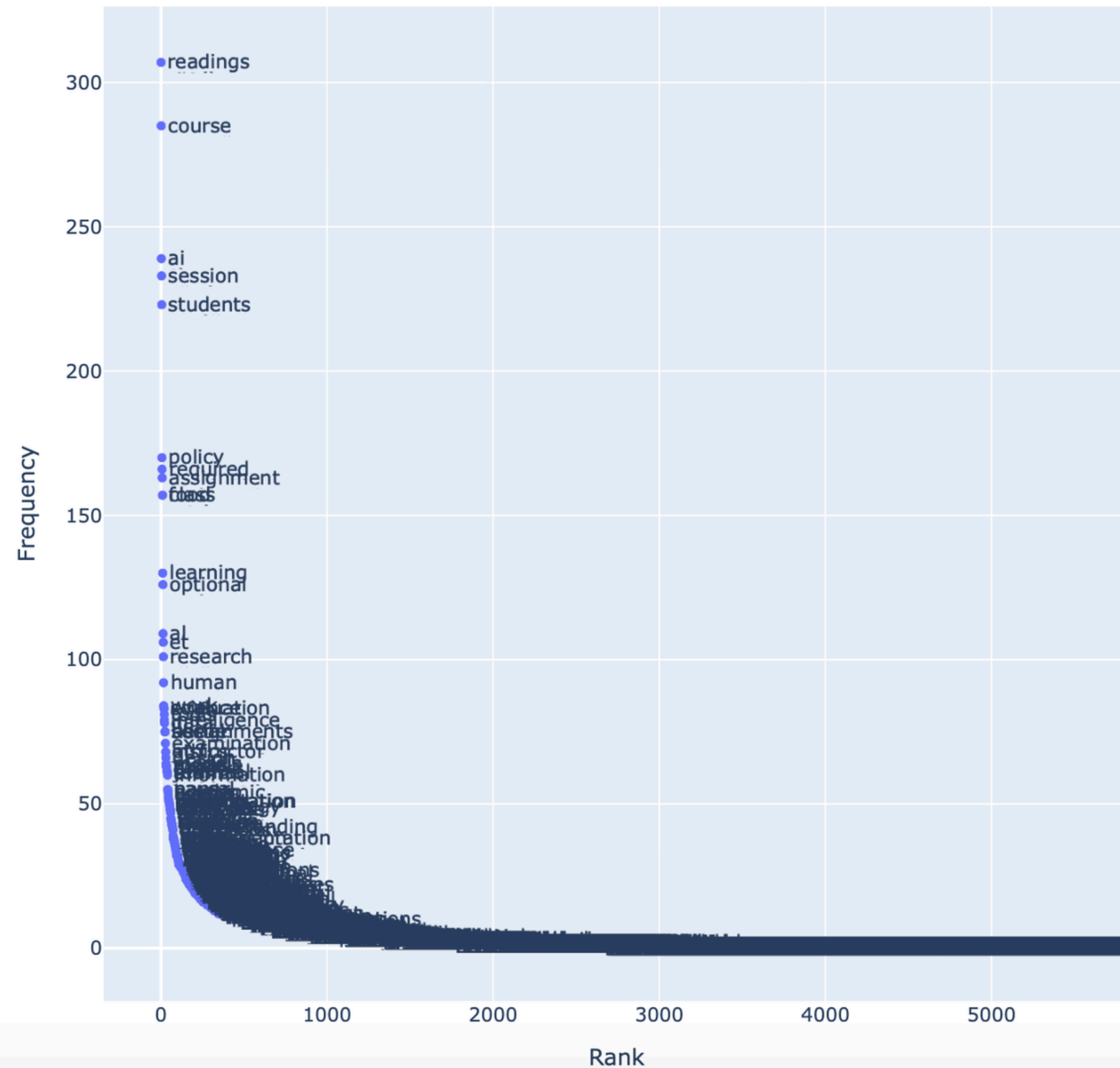
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2. **Inverse Document Frequency**: how rare is the term relative to the entire corpus?
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  - b.  $DF(t)$ : number of documents the term appears in

| word      | TF(t, d) | $N_d$ | DF(t) | IDF   | TF*IDF |
|-----------|----------|-------|-------|-------|--------|
| or        | 1        | 4     | 4     | $4/4$ | 1      |
| biases    | 1        | 4     | 1     | $4/1$ | 4      |
| platforms | 2        | 4     | 1     | $4/1$ | 8      |

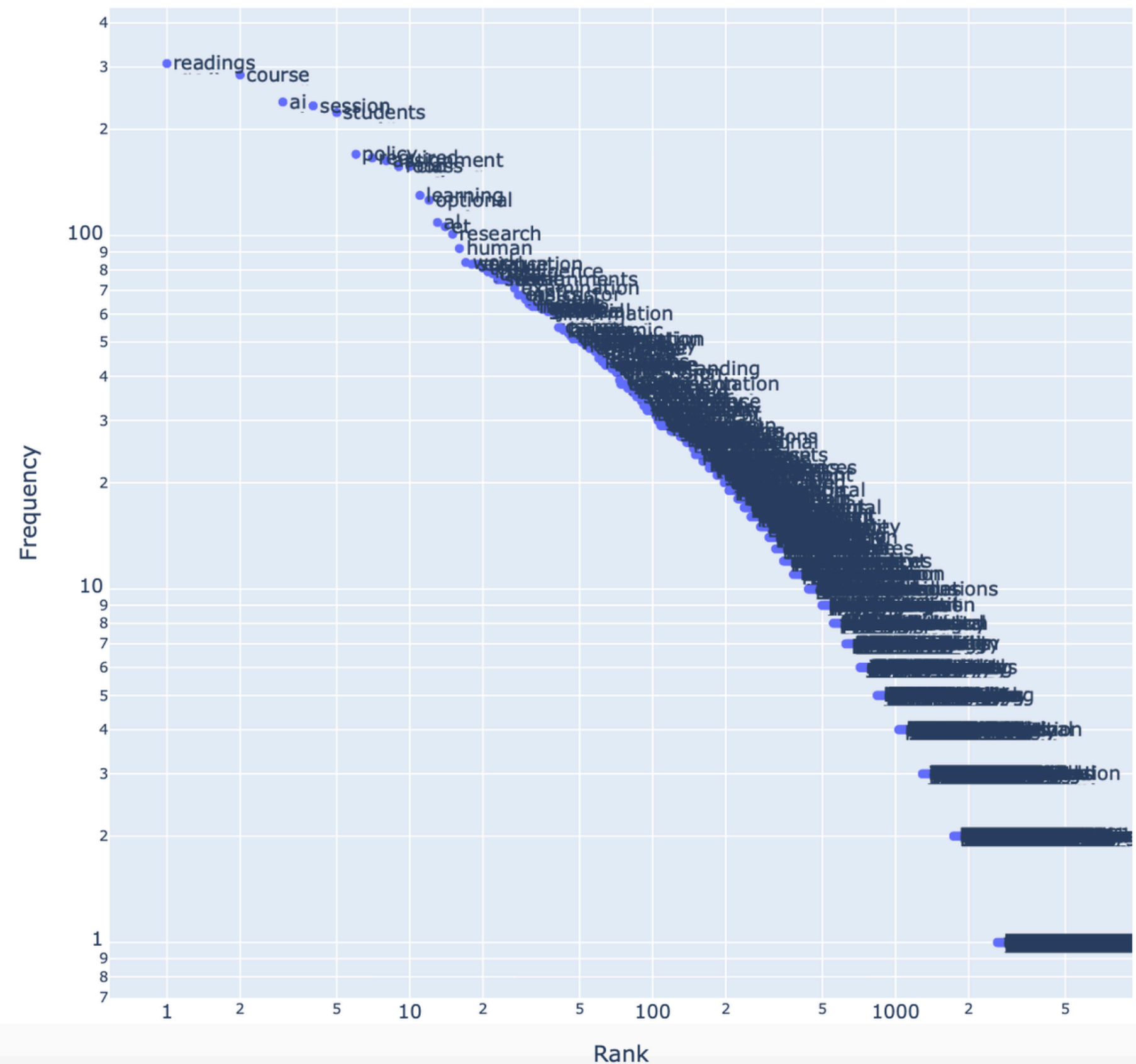
# TF-IDF and Zipf's Law



# TF-IDF and Zipf's Law

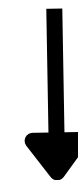


## log-log plot



# TF-IDF: Math

$$IDF = \ln \left( \frac{N_d}{DF(t)} \right)$$

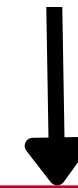


| word      | TF(t, d) | N_d | DF(t) | IDF  | TF*IDF |
|-----------|----------|-----|-------|------|--------|
| or        | 1        | 4   | 4     | 0    | 0      |
| biases    | 1        | 4   | 1     | 1.39 | 1.39   |
| platforms | 2        | 4   | 1     | 1.39 | 2.77   |

# TF-IDF: Math

scikit-learn implementation

$$IDF = \ln \left( \frac{1+N_d}{1+DF(t)} \right) + 1,$$



| word      | TF(t, d) | N_d | DF(t) | IDF  | TF*IDF |
|-----------|----------|-----|-------|------|--------|
| or        | 1        | 4   | 4     | 1    | 1      |
| biases    | 1        | 4   | 1     | 1.92 | 1.92   |
| platforms | 2        | 4   | 1     | 1.92 | 3.83   |

# Search Engine

## Natural Language Processing Lab 3

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</> Code ▼

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## 1 Objectives

Welcome to the third Natural Language Processing lab!

Today's objectives are to:

1.  Explore different **text representations** as vectors of token weights
2.  Calculate **cosine similarities** between documents.
3.  Build a **search engine** to compare a user query with corpus documents.

Next Week: Embeddings! 🧐

THANK  
YOU!